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Thermally Activated Condensation and Evaporation in Cylindrical Pores

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Abstract

In a confined geometry, thermal activation gives rise to specific condensation or evaporation mechanisms. Accounting for this phenomenon is therefore important for characterizing porous materials from their sorption isotherms. However, most phenomenological models discussing thermal activation in strong confinement are only qualitative. Here, we use a semi-macroscopic approach to compute the temperature and diameter dependences of the condensation and evaporation for a fluid confined in a cylindrical nanopore in presence of thermal activation. In particular, our model predicts the effect of confinement on cavitation. Our results show how thermal activation controls the existence and shape of the hysteresis loop in porous materials with non-interconnected cylindrical pores, and shed a new light on the applicability of the classical Barrett-Joyner-Halenda method for determining pore diameter distributions in porous materials. This work paves the way to a phenomenological multiscale approach able to describe condensation and evaporation in materials with connected pores.

1 Introduction

Condensation and evaporation of fluids in porous materials are described by sorption isotherms which differ from the bulk density-pressure isotherm in two respects. First, for wetting materials, both processes take place below the bulk saturated vapor pressure, the pressure shift increasing when the average pore diameter decreases. Second, these isotherms are generally hysteretic, condensation taking place at a larger pressure than evaporation does. The diameter dependence of the pressure shift is at the basis of the characterization of porous materials through their pore diameter distribution. For wide enough pores, typically larger than several tens of nanometers in diameter, it can be essentially understood in terms of capillarity and described by the Kelvin equation which states that the pressure shift is proportional to the liquid-gas fluid surface tension and to the inverse of the pore diameter. However, the Kelvin equation alone does not account for the existence of an hysteresis. For a perfect cylindrical pore open at both ends, it is now clear that the Kelvin equation gives the evaporation pressure, while the condensation process involves a capillar instability of the liquid film adsorbed on the pore walls.^{1,2} Furthermore, the attractive interaction between the substrate and the fluid which is responsible for the film formation also modifies the evaporation and condensation pressures with respect to their values in a pure capillarity approach, as pointed out a long time ago by a number of authors.³⁻⁶ Their results are at the basis of the scheme first proposed by Barrett, Joyner, and Halenda (BJH)⁷ to measure the pore diameter distribution of porous materials from the measurement of evaporation isotherms. However, such a scheme assumes that the pores are both ideal and not coupled, which most often is not the case. Moreover, it relies on a macroscopic approach, the validity of which may be questioned for nanometric pores.

Many experimental and theoretical studies have thus aimed to improve the fundamental understanding of condensation and evaporation in nanoporous materials, taking into account the microscopic nature of the problem (for reviews, see for example⁸⁻¹¹), the coupling between pores,¹²⁻¹⁴ or the existence of disorder.¹⁵⁻¹⁷ Today, the microscopic condensation and

evaporation processes are well understood in ordered porous materials, thanks to the combination of efficient theoretical techniques (density functional theory -DFT-, non linear DFT -NLDFT-, Grand Canonical Monte Carlo -GCMC-, ..)^{9-11,18} and of experiments performed on well-controlled mesoporous silica materials (e.g. MCM-41, SBA-15, SBA-16, ..)¹⁸⁻²⁰

The case of porous disordered materials is more complex, as relevant systems are too large to be accessible to direct molecular simulations. This limitation not only concerns disordered 3D porous glasses such as Vycor, but also systems of non-connected 1D pores in the presence of chemical or geometrical disorder along the pores length.^{17,21,22} The local mean field lattice DFT^{15,23} has been shown to reproduce remarkably well the experimental shape of the hysteresis loop and its evolution with temperature in disordered porous glasses such as aerogels^{24,25} or Vycor.^{11,16,26-28} However, it does not provide a simple picture in terms of basic mechanisms which could be used to discuss materials other than those simulated. An alternative is to use a multiscale approach as proposed in Ref. 17 to describe the effect of geometrical disorder on the hysteresis loop in linear pores. In this strategy, the porous material is modeled by an assembly of coupled pores. For a given chemical potential, the intra-pore state is obtained by GCMC with boundary conditions self-consistently set by the states of the pore's neighbours, hence the multiscale name. This scheme remains however heavy due to the use of GCMC.

A simpler multiscale strategy is at the heart of seminal works on evaporation in disordered porous materials.^{12-14,29} Here, the intra-pore behaviour is essentially modeled by the macroscopic Kelvin equation, which sets the diameter dependent pressure where the liquid and the gas phase can coexist within the pore. While this approach correctly grasps the percolative nature of evaporation, it is too crude for detailed comparison with actual experiments on nanoporous materials, since it does not take into the existence of a film. A modified Kelvin equation has been recently used by Morishige²² to discuss the case of corrugated pores viewed as a 1D chain of coupled pores. However, his approach assumes the film thickness to be independent on pressure, which is only an approximation.

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3 A better controlled starting point for a multiscale approach is the semi-macroscopic
4 approach developed by Saam and Cole,² which consider an ideal cylindrical pore open at
5 both ends on the vapor reservoir and a perfectly wetting substrate. Condensation first
6 proceeds by the adsorption of a film on the inner surface of the pore, due to the substrate
7 attraction. This film thickness increases with pressure, up to some critical pressure where
8 the pore fills irreversibly, due to a spinodal instability of the film (not to be confused with the
9 vapor bulk spinodal instability). In contrast, evaporation takes place reversibly by recession
10 of the vapor-liquid menisci present at the pore ends, leaving an adsorbed film behind. This
11 process occurs at an equilibrium pressure lower than the spinodal pressure, so that, between
12 these two pressures, the film is metastable with respect to the filled state. This scenario
13 qualitatively holds down to pores diameters of several nanometers, as confirmed by DFT³⁰
14 and GCMC simulations.³¹ A key ingredient is however missing in Saam and Cole's model,
15 and, more generally, in most other semi-macroscopic approaches, namely thermal activation.
16
17

18 Thermal activation has three effects. First, it favors condensation by allowing the pore
19 to fill below the spinodal pressure. Second, it plays a crucial role in the situation where the
20 pore is connected to the vapor reservoir through narrow constrictions (so-called ink-bottle
21 geometry). In this case, evaporation within the pore cannot take place at the equilibrium
22 pressure, since the constrictions are then filled with liquid, so that no meniscus preexists
23 at pore ends. In Saam and Cole's model, the pore would empty only once the constriction
24 does so, which takes place at its own, lower, equilibrium pressure, a situation coined pore-
25 blocking. In contrast, thermal activation may allow evaporation at an intermediate pressure,
26 by favoring the direct nucleation of a vapor bubble (*i.e.* cavitation) inside the pore.^{18,32–35}
27 Third, and a consequence of the two first effects, thermal activation makes the hysteresis
28 loop to close at a temperature lower than the bulk critical temperature,³² an alternative
29 mechanism to a confinement induced shift of the critical point.³⁰ Accounting for thermal
30 activation is thus an essential step towards a multiscale treatment of a disordered network
31 of cylindrical pores.
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In this paper, we extend Saam and Cole’s model to compute the energy barrier for activated condensation or evaporation, using an approach similar to that followed by Morishige and Tateishi³³ for the simpler case of spherical pores. Within this extended model, we compute the diameter and temperature dependences of the condensation and evaporation pressures. The results are in particular relevant for two pending issues in the field, namely the sensitivity of cavitation to confinement and the mechanism of hysteresis closure. We also clarify to which extent the pore diameter distribution of porous materials can be determined from the condensation or evaporation isotherms. For illustrative purpose, we will consider the case of helium confined in pores several nanometers in diameter, as an approximation of the case of helium in Vycor glass, which we previously studied experimentally.²⁸ However, we kept the presentation as general as possible so that our model can be used as a basis for a multiscale approach in other experimental systems.

2 Methods

Thermal activation has been previously taken into account by different theoretical methods, such as DFT and NLDFT,³⁶ GCMC,^{34,37,38} or mean field kinetic theory.³⁹ Here, we follow an alternate route by using a semi-macroscopic model to compute the relevant energy barriers. Although less exact than the numerical methods based on a microscopic description, this approach, being much simpler, is better suited to be extended to a multiscale description.

We consider an infinitely long cylindrical pore of radius R_0 , connected to a gas reservoir at the same temperature, the pressure P_g^∞ of which sets the chemical potential of the fluid. As Saam and Cole, we describe the confined fluid, either liquid or gas, by its bulk state equation, and assume the surface tension to have its bulk value (we neglect the curvature effect described by Tolman’s approximation⁴⁰). Interaction with the substrate is taken into account by an interaction potential $U(r)$, where r is the distance to the pore axis. The local chemical potential $\mu(r)$ is then equal to the sum of $U(r)$ and of the fluid bulk chemical

potential $\mu_b(r)$ ($b = l$ for the liquid phase, g for the gas one).

2.1 Equilibrium states of a cylindrical bubble

We first discuss the possible equilibrium states, restricting ourselves to those consistent with the pore's symetries, described by a cylindrical bubble of radius R (corresponding to a liquid film of thickness $R_0 - R$). For a given reservoir pressure, these states are given by the minima of the grand potential, under the constraint of an uniform local chemical potential $\mu(r) = \mu$. The volumic contribution to the grand potential is $\rho(r)(f_b(r) + U(r) - \mu)$ where $f_b(r)$ and $\rho(r)$ are the molar fluid bulk free energy and density, respectively. Introducing the local liquid and gas pressures $P_l(r)$ and $P_g(r)$, such that

$$\mu_l(P_l(r)) + U(r) = \mu_g(P_g(r)) + U(r) = \mu_g(P_g^\infty) = \mu \quad (1)$$

and using, for each phase, $\mu_b = f_b + P_b/\rho_b$, we obtain the difference of grand potential between the bubble and fully filled states. Per unit pore length, the grand potentiel referred to the filled state ($R=0$) is

$$\Omega(R) = \int_0^R (P_l(r) - P_g(r)) d^2r + 2\pi\sigma R \quad (2)$$

where σ is the fluid surface tension.

$\Omega(R)$ is then obtained by computing $P_l(r)$ and $P_g(r)$ from equation 1 and the equation of state (EoS) relating μ_b to P_b in each phase, and inserting the results in equation 2. This procedure requires a detailed knowledge of the fluid EoS. General results can however be obtained by assuming that $P_g(r)$ and $P_l(r)$ are close enough to the bulk saturated vapor pressure P_{sat} to linearly expand the chemical potential with respect to pressure. This assumption, which amounts to neglect the liquid and gas compressibilities, gives

$$\Omega(R) = \int_0^R \Delta\rho(\Delta\mu - U(r)) d^2r + 2\pi\sigma R \quad (3)$$

where $\Delta\rho = \rho_l(P_{\text{sat}}) - \rho_g(P_{\text{sat}})$ and $\Delta\mu = \mu_g(P_g^\infty) - \mu_g(P_{\text{sat}})$.

This result remains valid when only the liquid compressibility is neglected, provided the gas density is negligible with respect to that of the liquid, as assumed in Ref. 2. When the temperature T approaches the bulk critical temperature T_c , the density difference between the gas and liquid vanishes, and their compressibilities diverge, so that eq.3 should fail. However, this only occurs very close to T_c . For example, in the case of helium in a silica pore, we show in the Supporting Information that the predictions of eq.3 hold to within 1% of T_c . We therefore use eq.3 as the basis of our analysis.

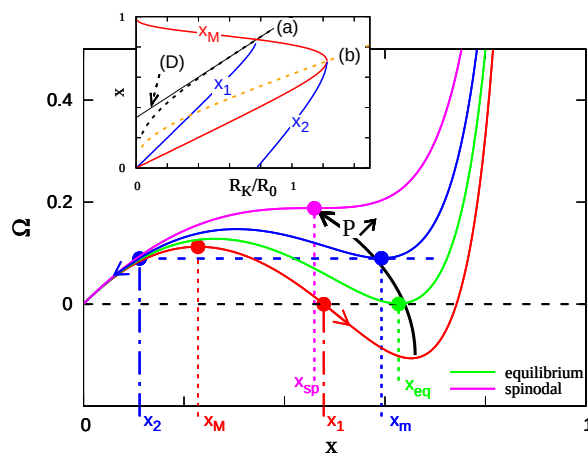


Figure 1: Typical grand potential Ω of a cylindrical bubble of reduced radius $x = R/R_0$ for increasing pressures at a given attractive potential. The characteristic reduced radii are defined in the text. Arrows indicate the direction of spontaneous expansion or collapse. Inset: Pressure dependence of the characteristic radii for a van der Waals potential corresponding to a normalized attraction strength $R^*/R_0 = 0.2$. The pressure is parametrized by the adimensional ratio R_K/R_0 . Dashed lines (a) and (b) are the loci of the equilibrium and spinodal radii when R^*/R_0 is varied. The straight line (D) corresponds to Derjaguin's expression for the equilibrium radius.

The typical behaviour of $\Omega(R)$ for an attracting potential $U(R)$ is shown in Figure 1. At low R , $\Omega(R)$ increases linearly due to the surface term. At large R , it also increases due to $U(R)$ which favors the liquid phase near the walls. The intermediate behaviour depends on the undersaturation $\Delta\mu$. For negative enough $\Delta\mu$, $\Omega(R)$ is non monotonic and presents a maximum at a reduced radius x_M and a minimum at x_m , corresponding to a liquid film covering the pore walls. For a specific value of $\Delta\mu$, corresponding to a

reservoir pressure P_{eq} and a minimum reduced radius $x_{\text{m}} = x_{\text{eq}}$, the film has the same potential as the filled state. The film is absolutely stable below P_{eq} and metastable above, up to a spinodal pressure P_{sp} , where the minimum disappears ($x_{\text{m}} = x_{\text{sp}}$). In the absence of thermal activation, condensation takes place abruptly at P_{sp} , while evaporation takes place at $P_{\text{g}}^{\infty} = P_{\text{eq}}$ by recession of menisci. Figure 1 shows two other characteristic radii important for discussing the effect of thermal activation: x_1 , such that $\Omega(x_1) = \Omega(0)$, and x_2 , such that $\Omega(x_2) = \Omega(x_{\text{m}})$. Below equilibrium, a cylindrical bubble of radius larger than x_1 spontaneously expands to x_{m} . Above, a bubble of radius smaller than x_2 spontaneously collapses to the filled state.

As Saam and Cole, we restrict ourselves to the case of a Van der Waals interaction between the fluid and the substrate.⁴³ For helium on silica, this is supported by precise measurements of adsorption on alkaline-earth fluoride crystals by Sabisky and Anderson, which show that the Lifshitz theory of Van der Waals forces holds in the range 1-25 nm.⁴⁴ For a cylindrical pore, this choice of a van der Waals interaction leads to²

$$U(r) = -\frac{3\pi\alpha}{2R_0^3} F\left[\frac{3}{2}, \frac{5}{2}; 1; \left(\frac{r}{R_0}\right)^2\right] \quad (4)$$

where F is a hypergeometric function (satisfying $F(0)=1$) and α the interaction strength, such that the interaction between a half space filled with silica and a helium atom at a distance z from the silica interface is given by $-\frac{\alpha}{z^3}$. For a distance to the wall much smaller than the pore radius, $U(z)$ reduces to the latter expression, which is the limit considered by Derjaguin.⁴

This choice of a van der Waals interaction allows to fully describe the problem in terms of three lengths. The first one, the pore radius R_0 is purely geometrical. The second length, R_{K} , is given by:

$$R_{\text{K}} = -\frac{2\sigma}{\Delta\mu\Delta\rho} \quad (5)$$

R_{K} is the radius of the spherical critical germ for bulk cavitation, that is the inverse of the

mean curvature of a free interface ($U=0$) in mechanical equilibrium at $\Delta\mu$. R_K parametrizes the undersaturation $\Delta\mu$, *i.e.*, for a given EoS, the pressure.

The last length, R^* , introduced by Saam and Cole², measures the confinement through the relative strength of wall attraction compared to capillary effects :

$$(R^*)^2 = \frac{3\pi\alpha\Delta\rho}{\sigma} \quad (6)$$

With these definitions, eq 2 is recast in the adimensionnal form:

$$\frac{\Omega(x) - \Omega(0)}{2\pi\sigma R_0} = x - x^2 \frac{R_0}{R_K} + \frac{R^{*2}}{R_0^2} \int_0^x y f(y) dy \quad (7)$$

with $x = R/R_0$ and $f(x) = \frac{1}{2}F\left[\frac{3}{2}, \frac{5}{2}; 1; x^2\right]$.

Equation 7 fully characterizes the equilibrium states of a cylindrical bubble. It allows to derive the equations obeyed by the characteristic radii x_m , x_M , and x_1 or x_2 (see Supporting Information). Solving numerically these equations give the dependence of these radii on the undersaturation R_K/R_0 , as illustrated in the inset of Figure 1 for $R^*/R_0=0.2$. The inset also shows the loci of $(R_{K,eq}/R_0, x_{eq})$ and of $(R_{K,sp}/R_0, x_{sp})$ when R^*/R_0 varies. As obvious from Figure 1, $x_m = x_1 = x_{eq}$, and $x_2 = 0$ at the equilibrium pressure, and x_m , x_M and x_2 merge at x_{sp} at the spinodal pressure.

The values of R_K/R_0 determining the equilibrium and spinodal pressures can be determined from x_{eq} , x_{sp} , and eq.7 (see eq.S1 of Supporting Information). They are plotted in Figure 2 as a function of R^*/R_0 . At low R^*/R_0 (small attraction, large surface tension, or large pore radius), they approach the results of the purely capillary approach ($U=0$), namely $R_K = R_0$ at equilibrium, and $R_K = 2R_0$ at the spinodal point.⁴⁵ In the opposite limit of large R^*/R_0 (reached for strong attraction, small surface tension, or small pore radius), the substrate attraction becomes the dominant effect, and the equilibrium and spinodal values of R_K/R_0 tend to zero. In this limit, capillarity does not come into play, and the pore fills up continuously to $x = 0$, reached at $P_g(0) = P_{sat}$ or, equivalently, $\Delta\mu = U(0)$, corresponding

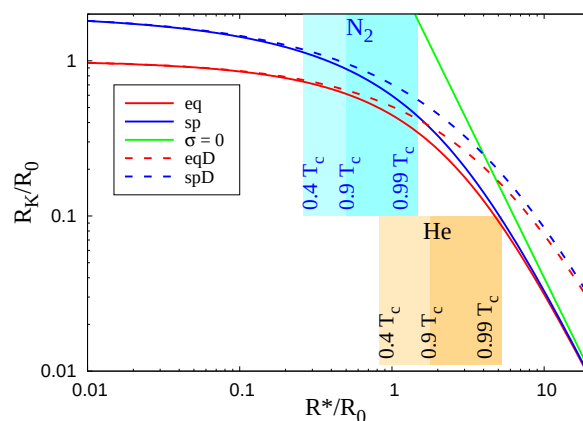


Figure 2: Dependence of the equilibrium and spinodal pressures, measured by R_K/R_0 , on the normalized interaction strength R^*/R_0 . The shaded areas give the range of R^*/R_0 values for helium and nitrogen in a 4 nm radius pore, and temperatures ranging from $0.6 T_c$ to $0.99 T_c$. Dashed lines correspond to Derjaguin's approximation.

to $R_K/R_0 = 4(R_0/R^*)^2$. Between these two limits, capillarity and attraction play a role. The deviation of the equilibrium and spinodal values of R_K/R_0 from the capillary approach, corresponding to larger deviations from P_{sat} , can be qualitatively understood in terms of the reduction of the pore diameter due to the finite film thickness. For $R^*/R_0 < 2$, R_K/R_0 at equilibrium is larger than 0.25. In this range, the equilibrium line shown in Figure 1 is well approximated by $x_m = 1/3 + 2/3 R_K/R_0$, *i.e.* $R_K = R_0 - 3/2t$, where t is the film thickness. The equilibrium value of R_K is then given by Kelvins's approach, with an effective pore radius equal to R_0 minus three half the film thickness. This corresponds to Derjaguin's result for a van der Waals interaction.⁴ Note however that Derjaguin makes the approximation of a planar substrate. Due to the substrate curvature, this approximation underestimates t , so that the corresponding condensation pressure underestimates the deviation from Kelvin's approach by more than 10% for $R^*/R_0 = 1$. Figure 2 shows that, for a 4 nm radius pore, this effect of the pore curvature is small for nitrogen below 100 K, but not for helium above 4 K, resulting from the relatively smaller ratio of surface tension to liquid density in the latter case.

2.2 Thermally activated nucleation

Without thermal activation, hysteresis disappears when the surface tension vanishes, and the spinodal and equilibrium pressures coincide. In Saam and Cole's model, this happens at the bulk critical point. When thermal activation is taken into account, the hysteresis loop closes at a lower temperature, T_{ch} , such that the energy barrier separating at equilibrium the two stable states can be overcome by thermal activation. This effect, first pointed out by Machin,³² has been studied by Morishige and Tateishi in the case of a spherical pore.³³ For a cylindrical pore, the situation is more complex as the maximum of $\Omega(R)$ in equation 7 is extensive with the pore length, i.e. diverges for an infinitely long pore. Liquid nucleation requires to break the translational symmetry and to identify the nucleation path which minimizes the energy barrier. For $U(r) = 0$, this problem has been considered for vapor nucleation inside an hydrophobic cylindrical pore by Lefevre *et al.*⁴⁶ In the same $U(r) = 0$ limit, the problem of the film stability versus axisymmetric perturbations has been exactly studied by Everett and Haynes for a perfectly wetting cylindrical pore.⁴⁷

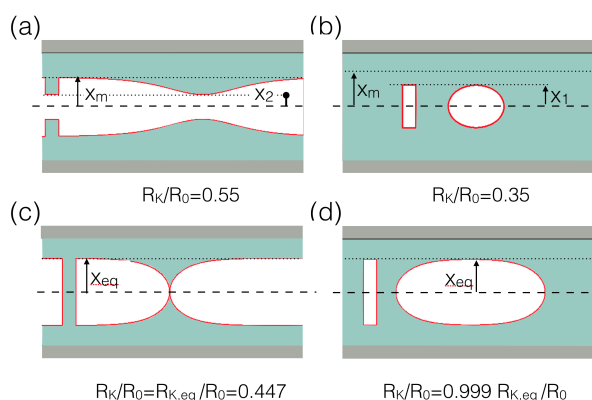


Figure 3: Critical germs for a normalized interaction strength $R^*/R_0=1$: (a) critical bump for nucleation above equilibrium; (b) critical bubble for cavitation below equilibrium; (c) critical bump for nucleation at equilibrium; (d) critical bubble for cavitation just below equilibrium. In each case, the exact critical profile is represented on the right and the disk approximation on the left.

Here, we go beyond these previous works by studying thermally activated condensation and evaporation in nanoscale pores where $U(r)$ cannot be neglected. As discussed above,

a cylindrical bubble of radius larger than x_1 spontaneously expands, both laterally and radially, up to the radius x_m . In the opposite case, a cylindrical bubble of radius less than x_2 spontaneously collapses to fill the pore. Therefore, the energy barrier to overcome is given by the minimal energy of a deformation (bubble or bump) allowing to reach x_1 for cavitation or x_2 for activated condensation. The process is illustrated on the left of each panel of Figure 3 assuming the critical germ to be an infinitely thin disk of radius x_1 (cavitation) or a thin annulus between x_m and x_2 (liquid nucleation). In this case, the germ energy is purely surfacic and equals $2\pi\sigma x_1^2 R_0^2$ for cavitation, and $2\pi\sigma(x_m^2 - x_2^2)R_0^2$ for liquid nucleation. From Figure 1, it is clear that, as physically required, both values are equal at equilibrium, and that the barrier for liquid nucleation vanishes at the spinodal point. This provide an upper bound for the activation barrier (shown as thin lines in Figure 4a). In order to improve upon this bound, we have to balance the surface versus volume energy by optimizing the deformation profile to minimize the total germ energy. This procedure is carried out exactly in the Supporting Information where we show that $K(r)$, the adimensional local total curvature of the critical bubble or bump obeys

$$\sigma R_0 K(r) = P_g(r) - P_l(r) \quad (8)$$

where $P_g(r)$ and $P_l(r)$ are given by eq.1. eq.8 is nothing else than the Laplace equation expressing the local mechanical equilibrium of the interface. The profile of the critical bump or bubble is found by solving this equation for axisymmetric germs with the appropriate boundary conditions. As detailed in the Supporting Information, the critical bump above equilibrium is found by integrating eq.8 starting at reduced radius x_2 with an horizontal slope (Figure 3a). As required, the corresponding profile asymptotically tends to x_m , the film reduced radius. Below equilibrium, the critical bubble is found by integrating the same equation, starting on the pore axis with a vertical slope, up to x_1 which is reached with an horizontal slope (Figure 3b). At equilibrium, the critical profile coincides with the

equilibrium meniscus shape computed by Saam and Cole,² and is identical for cavitation and condensation (Figure 3c). Note that, because x_2 vanishes in this case, the slope of the bump profile at x_2 is singular instead of zero above equilibrium. On the cavitation side, continuity with equilibrium is ensured by the fact that the lateral extent of the critical bubble diverges as $R_{K,eq}$ is approached from below (Figure 3d).

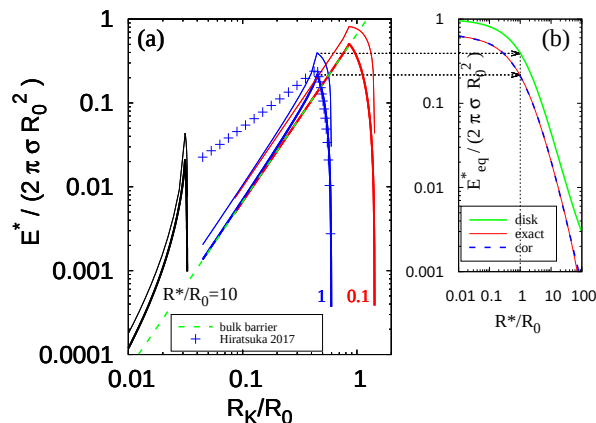


Figure 4: (a) Undersaturation dependence of the critical germ energy for cavitation (below maximum) and liquid nucleation (above maximum) for three adsorption strengths ($R^*/R_0 = 0.1, 1, 10$). (b) Dependence of the critical energy at equilibrium on the adsorption strength. In both figures, thick lines give the exact germ energy, and thin lines the energy in the disk approximation (see text). The crosses correspond to the prediction of Hiratsuka *et al.*⁴⁸

From the germ profile, we calculate the associated grand potential, hence the nucleation barrier E^* . Once normalized by the characteristic surface energy $2\pi\sigma R_0^2$, this barrier is a sole function of the adimensional pressure R_K/R_0 and adsorption strength R^*/R_0 .

3 Results and discussion

3.1 Energy barrier

Thick lines in Figure 4a show how the energy barrier depends on R_K/R_0 for different R^*/R_0 . For each curve, the barrier is maximal at the R_K/R_0 value corresponding to equilibrium ($R_K = R_{K,eq}$) and vanishes at the spinodal pressure ($R_K = R_{K,sp}$). Below $R_{K,eq}$, the curve

gives the critical energy for cavitation, and, above, for liquid nucleation. Well below $R_{K,eq}$, the energy barrier for cavitation is close to the bulk value $4/3\pi\sigma R_K^2$. This corresponds to a regime where the critical bubble is spherical, of radius much smaller than the pore radius R_0 . The deviation with respect to the bulk barrier is due to the finite value of $U(0)$ (see eq.S19 of Supporting Information). When R_K approaches equilibrium, the critical bubble is deformed due to the walls potential which favors the liquid phase (Figure 3d) and the barrier strongly increases with respect to the bulk case. As physically expected, the relative deviation increases with increasing adsorption strength. Note that the deviation on the cavitation side occurs for R_K smaller than R_0 . In other words, confinement affects cavitation even for a pore radius larger than the radius of the bulk critical germ. For a large attraction strength, this is not a marginal effect (*e.g.*, for $R^*/R_0 = 10$, confinement has a large influence on cavitation for a pore thirty times larger than the radius of the bulk critical germ). This results from the long range nature of the van der Waals potential.

The behaviour found for $E^*(R_K/R_0)$ is remarkably similar⁴⁹ to that found by Vishnyakov and Neimark for nucleation in nanoscale capillaries using microscopic Monte Carlo simulations.³⁸ This demonstrates that our semi-macroscopic model properly captures the physics of nucleation in nanopores. Not too close to equilibrium, the cavitation barrier is, within 10%, 2/3 of the value found in the disk approximation.

Our result can also be compared to a recent paper by Hiratsuka *et al.*,⁴⁸ who developed a model of activated condensation in a cylindrical open-ended pore. This model is similar to that of Saam and Cole except that the authors also include a curvature dependence of the surface tension. In contrast to the present work, the authors do not compute the true barrier, but estimate its value as the energy difference between the minimum and the maximum of the grand potential of a cylindrical bubble. As we stressed above, the latter quantity is extensive in length. The authors then get the barrier by assuming the length of the critical germ to be equal to the pore diameter. As shown by Figure 4a (see also Figure S3 of the Supporting Information), their result matches our prediction. This results from a cancellation between

two sources of error in their calculation (the neglect of the surface energy associated with the lateral walls of their germ, and the non optimization of the germ shape). In our view, this match explains the reported good agreement between their prediction and their results of gauge cell Monte Carlo calculations. However, using the same *ansatz* to compute the barrier energy for cavitation results in much too large barrier energies (Figures 4a and S3). This shows the limitations of the model of Ref. 48.

A quantity of specific interest is E_{eq}^* , the normalized barrier height at equilibrium, which, as discussed below, sets the critical temperature for hysteresis closure. This quantity is shown in Figure 4b. For a vanishing attractive potential ($R^*/R_0 = 0$), $E_{eq}^* = 4/3\pi\sigma R_0^2$, the activation energy for bulk cavitation at the equilibrium pressure (since, at equilibrium, $R_K = R_0$ for $R^*/R_0 = 0$). As the potential increases, the barrier decreases. In the range $0.1 < R^*/R_0 < 10$, it is well approximated by multiplying the disk approximation result (x_{eq}^2) by a correcting factor $0.54 (R^*/R_0)^{-0.073}$ (Figure 4b). This provides a quick way to evaluate the effect of thermal activation for any system where the fluid interacts with the substrate through a van der Waals potential.

As a final point, we stress that our model only considers homogeneous cavitation, and does not include the possibility of heterogeneous cavitation on the pore walls. For a perfectly wetting fluid, and perfectly smooth walls, it is indeed expected that the barrier for heterogeneous cavitation is larger than for homogeneous cavitation. Whether, in actual materials, the walls roughness may change this picture remains an open question.

3.2 Condensation and evaporation pressures

Our model allows to describe the effect of confinement on the activated condensation and evaporation pressures. These pressures are determined by the critical barrier that thermal activation allows to overcome. The Arrhenius exponential dependence of the transition probability on the energy barrier translates to a weak logarithmic dependence of the critical barrier height on the observation timescale and the experimental volume. As a result, the

dimensionless critical barrier height for nucleation $N_c = \Omega/k_bT$ weakly depends on the experimental system considered. In the following, we will thus consider N_c to be a constant irrespective of the temperature and pore radius. Unlike for equilibrium and spinodal instability, the critical values of R_K/R_0 for activated nucleation or cavitation do not only depend on R^*/R_0 , but also, due to the normalization of the energy barrier in Figure 4, on the fluid surface tension and the pore radius. For the sake of discussion, we will mainly consider the case of helium, taking $N_c = 40$ from bulk helium cavitation measured by ultrasonic experiments.⁵⁰ In order to avoid the fluid dependent conversion from $\Delta\mu$ to pressure, we will express $\Delta\mu$ as a fugacity. In this way, obtaining results for any van der Waals fluid is straightforward following the same lines as for helium.

For given pore radius, temperature, and fluid surface tension, the activated condensation and evaporation fugacities are obtained from the normalized barrier energy at the appropriate R^*/R_0 value. If the maximum of this curve is smaller than $F(R_0, T) = N_c k_b T / (2\pi\sigma(T)R_0^2)$, the energy barrier can be overcome at equilibrium and condensation and evaporation are non hysteretic. In the reverse case, the activated evaporation or condensation fugacities are obtained from the two intercepts of the curve with $F(R_0, T)$. These two regimes respectively occur above/below the pore-dependent critical temperature $T_{ch}(R_0)$ such that the maximal normalized germ energy equals $F(R_0, T)$.

Figure 5 illustrates how, at a given temperature, the characteristic fugacities vary with the pore radius in the range 3 nm (a typical value for Vycor pores) up to 20 nm (a representative value for porous alumina⁵¹). The activated fugacities are computed as just described. The equilibrium and spinodal fugacities are found from Figure 2 by converting R_K values to fugacities using eq.5. These fugacities decrease with the pore radius due to the increase of both van der Waals and capillary effects. Below the critical radius R_c such that $T_{ch}(R_c) = T$, the equilibrium and activated fugacities coincide, and condensation/evaporation is reversible. This finding agrees with the results of Monte Carlo simulations and NLDFT.^{19,52} Above R_c , the condensation and evaporation fugacities depart from the equilibrium line. For large R_0 's

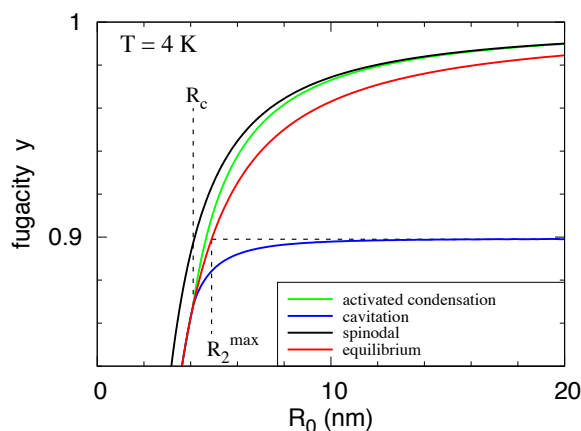


Figure 5: Pore radius dependence of the characteristic fugacities for the helium/silica system at $T=4$ K ($R^*=5$ nm). The condensation/evaporation process is non hysteretic below the critical radius $R_c(T)$. $R_2^{\max}$ is the maximal constriction radius to observe cavitation in an ink-bottle geometry with a large cavity.

values, the condensation barrier becomes too large to be overcome by thermal fluctuations, and the condensation fugacity tends to the spinodal fugacity. In contrast, the cavitation barrier and fugacity tend to their bulk limits. The transition from one regime to the other takes place over some radius range above R_c . In this range, the cavitation fugacity decreases when the pore diameter is decreased, *i.e. confinement inhibits cavitation*.

Let us give two examples of how the pore diameter dependence predicted by our model can be used. The first one is the ink-bottle geometry where a cylindrical cavity of radius R_1 is connected to the vapor reservoir through a cylindrical constriction of smaller radius R_2 . Practical realizations of this situation are arrays of ink-bottles^{53–57} and cagelike mesoporous materials consisting in networks of cavities connected by narrow constrictions.^{18,34,58} Figure 5 allows to predict the conditions to observe cavitation in these geometries. At a given temperature, the cavity empties by cavitation before the vapor-liquid meniscus recesses into the constriction if $y_{\text{cav}}(R_1) > y_{\text{eq}}(R_2)$. In the opposite situation of pore-blocking, the cavity only empties when the constriction does. For a large cavity, the largest constriction radius to observe cavitation is $R_2^{\max} \gtrsim R_c$ and approaches R_c as R_1 tends to R_c . We have applied our model to compute the maximal constriction diameter for observing cavitation in nitrogen

and argon at 77 K. Using the same value $N_c = 40$ as for helium and the physical parameters for these fluids (see Supporting Information for values), we find $R_2^{\max}$ to be 3.6 nm in nitrogen and 2.8 nm in argon. Therefore, cavitation should be easier to observe using nitrogen rather than argon. Reversely, evaporation from cage-like materials can probe the constriction diameter down to smaller sizes when using argon rather than nitrogen. This conclusion agrees with the results of Ref. 58. The value of 3.6 nm is also close to the 4 nm quoted in the study of Ref. 34 as the maximal constriction diameter in various systems where nitrogen cavitation is observed. We thus believe that our model should be able to explain why cavitation is observed in some experiments on ink-bottle arrays,^{53,54} but not in others.⁵⁵⁻⁵⁷

The second example concerns systems of non-interconnected cylindrical pores of nanometric diameter, such as porous silicon,⁵⁹ silica MCM-41 templated pores⁶⁰⁻⁶² or alumina nanopores.^{57,63,64} Discrepancies with the behaviour expected for ideal pores^{51,57,59,64} have been theoretically explained in terms of defects along the pores length, either chemical or geometrical.^{65,66} While this hypothesis is qualitatively supported by experimental and GCMC studies performed on modulated pores,⁶⁷ a multiscale approach based on our model could be used to quantify whether the existing observations can be explained by a realistic variation of the pores cross-section along their axis.

Let us now discuss the temperature dependence, for a given pore diameter, of the characteristic fugacities. The resulting temperature-pressure "phase diagram", shown in Figure 6 for helium and a pore radius $R_0 = 3$ nm, is the equivalent for a cylindrical pore of the diagram previously computed by Morishige and Tateishi for a spherical one.³³ Below the critical temperature $T_{ch}(R_0)$, the activated condensation and evaporation lines define a region within which the pore can be either filled or in the film state, depending on history and geometry. Above the condensation line, a pore open at both ends fills by activated condensation. Below the evaporation line, a pore closed by small enough constrictions empties by cavitation. These two lines merge with the equilibrium line at the critical temperature $T_{ch}(R_0)$, above

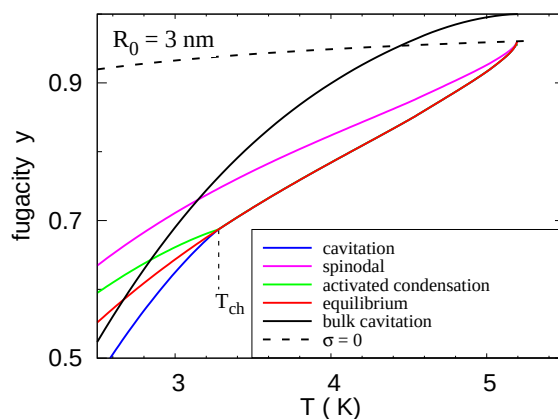


Figure 6: Phase diagram temperature T -fugacity y for helium in a 3 nm radius cylindrical pore. The lines corresponding to the activated condensation and evaporation fugacities merge with the equilibrium one at the critical temperature $T_{ch}(R_0)$, above which the condensation/evaporation process becomes reversible.

which thermal activation makes condensation and evaporation reversible. Above $T_{ch}(R_0)$, the equilibrium and spinodal y 's increase with temperature due to the surface tension decrease. At the bulk critical temperature T_c , where the surface tension vanishes, these two fugacities are smaller than unity (*i.e.* the pressure remains below the saturated vapor pressure) due to the van der Waals interaction. Their common value is equal to the condensation fugacity computed in the absence of surface tension (curve $\sigma=0$ in Figure 6), such that $P_g(0)$, the local pressure on the pore axis, is equal to the saturation pressure P_{sat} . This corresponds to $y(T) = \exp[-|U(0)|/k_b T]$.

Figure 6 shows that, in contrast to the activated condensation and cavitation pressures, the equilibrium pressure below T_{ch} smoothly extrapolates to the equilibrium pressure above. This supports the method proposed by Morishige and Nakamura⁶⁸ to identify, in complex porous materials, which of the condensation or evaporation branches below T_{ch} takes place at equilibrium.^{33,68–70}

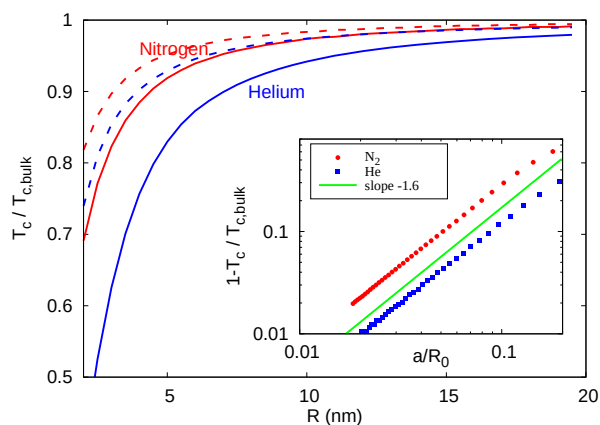


Figure 7: Predicted radius dependence of T_{ch} , the critical temperature for hysteresis, for helium and nitrogen. Full lines correspond to the present exact calculation, while dashed lines correspond to Machin's limit (no attraction, *i.e.* $R^*=0$). The inset shows that the relative shift from the bulk critical temperature varies as a power of a/R , where a is the average interatomic distance in the liquid phase far below T_c . The line corresponds to an exponent -1.6.

3.3 Critical temperature for hysteresis

The temperature dependence of R_c , or, equivalently, the radius dependence of T_{ch} , is determined by the condition that E_{eq}^* , the nucleation barrier at equilibrium, equals $N_c k_b T$. Using the above computation of E_{eq}^* and eq.6 to obtain this barrier as a function of T and of R_0 , we obtain $T_{\text{ch}}(R_0)$ for given fluid, substrate, and N_c . Figure 7 illustrates the result for helium and nitrogen (see Supporting Information for the physical parameters used). In both cases, confinement lowers T_{ch} , hence *favors the effect of thermal activation*. As discussed in the Supporting Information, this does not contradict our previous conclusion that confinement decreases the cavitation pressure. T_{ch} approaches the bulk critical temperature T_c only for large R_0 's values, above 20 nm in our examples. Also shown in Figure 7 is the predicted critical temperature when replacing E_{eq}^* by its value at zero R^* . It is easy to show that, at this temperature, the bulk cavitation pressure equals the Kelvin equilibrium pressure in the pore of radius R_0 . This is precisely the assumption made in Machin's original approach.³² As shown by Figure 7, the $E_{\text{eq}}^*(R^*)$ dependence significantly decreases T_{ch} with respect to Machin's prediction.

An alternative mechanism for an hysteresis closure at a temperature smaller than T_c is a shift of the critical temperature due to confinement. In this case, the surface tension vanishes, hence the hysteresis closes, at the shifted critical temperature, T_{cc} . The relative difference of T_{cc} with respect to T_c is expected to vary as a power law of the pore radius, with an exponent between -1 and -2 depending on the distance to T_c .³⁰ As shown by the inset of Figure 7, the thermal activation mechanism for T_{ch} also predicts an effective power law behaviour for the relative shift of T_{ch} from T_c with an exponent in this range (around -1.6, see Supporting Information for the origin of this value). Hence, the mere observation of a power law behaviour of the hysteresis closure temperature cannot prove that the critical temperature is shifted by the confinement. On the other hand, an exponent around -1 has been reported for the closure temperature of MCM-41 cylindrical pores with different pores diameters,⁷¹ and reproduced by GCMC calculations.⁷² While such an exponent is not reproduced by our model, thermal activation may also contribute to the closure of hysteresis in this case.

3.4 Isotherms

We now use our model to compute sorption isotherms, first, for a single pore, second, for an assembly of independent pores with distributed diameters. The results will allow us (i) to clarify the relationship between the isotherms shape and the underlying condensation and evaporation processes; (ii) to discuss the adequacy of the Barrett-Joyner-Halenda (BJH) method for extracting a pore diameter distribution from isotherms measurements.

For a single pore, and in an incompressible approximation, the average density in the pore is $\rho = (1 - x_m^2)\rho_l + x_m^2\rho_g(T, P_\infty)$. However, for temperatures above T_c or for small enough pores, this approximation does not hold, and one has to integrate over the pore radius the local liquid and vapor densities. Using eq.1 to compute the local pressures P_l and P_g , and the bulk EoS to convert these pressures into densities,⁷⁴ we compute the fluid density, averaged over the pore volume, in the film and filled states. The isotherms are then

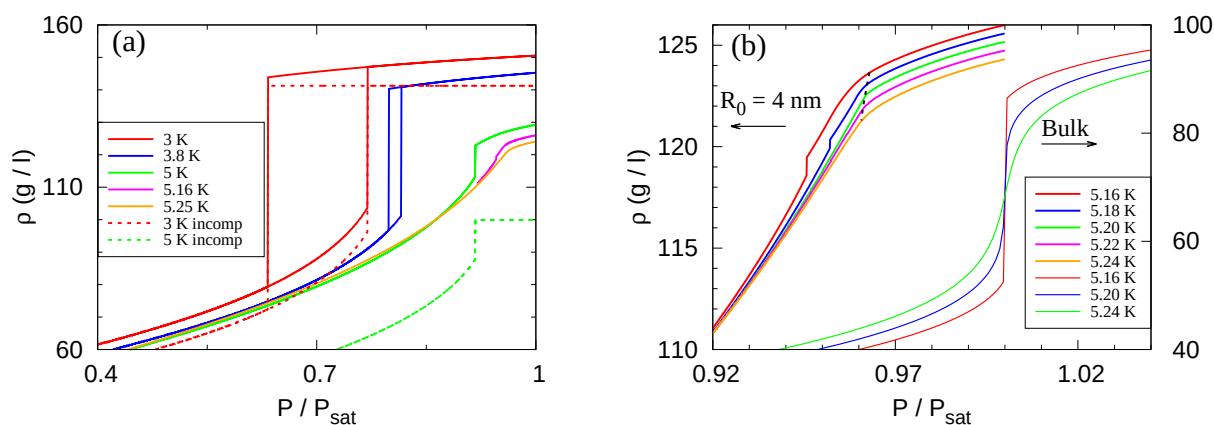


Figure 8: (a) Predicted helium isotherms for a 4 nm radius perfect cylindrical pore open at both ends. The dashed lines correspond to the incompressible approximation for the liquid. (b) Behaviour close to T_c and corresponding bulk isotherms.⁷³ The dashed line is the locus of points such that the on-axis pressure equals the bulk saturated pressure below T_c , and the pressure of maximal compressibility above.

determined, once the limits of stability of the liquid and film states are chosen. For a perfect cylindrical pore open at both ends, the pore fills at the activated condensation pressure, and empties at the equilibrium pressure. Figure 8a shows the corresponding isotherms for a pore radius 4 nm and temperatures below T_c (well below, slightly below, and above $T_{ch} \approx 4$ K) or slightly (1%) above.⁷⁵

Comparing the isotherms at 3 K and 5 K to those computed within the incompressible approximation shows that the liquid compressibility significantly affects the average density, both in the film and liquid states. In the film state, the effect increases with temperature, reflecting the increased liquid compressibility. In the liquid state, at the lower temperatures, the slope of the isotherm is found much larger than that expected from the bulk compressibility. The origin of this behaviour is explained in the Supporting Information.

Figure 8b shows that confinement drastically affects the isotherms around the bulk critical point. Below T_c , the density jump is much smaller than in the bulk case due to the large liquid contribution to the film state density. Above T_c , the divergence of the bulk compressibility is washed out by the spatial integration, only leaving a rounded kink of the density at the reservoir pressure such that the on-axis pressure is P_{max} , the pressure where the bulk

compressibility is maximal. This kink persists below T_c , now corresponding to an on-axis pressure equal to P_{sat} .⁷⁶ The important conclusion to be drawn from Figure 8b is that, even for a monodisperse collection of perfect pores, confinement makes it difficult to locate the precise critical temperature using isotherms measurements.

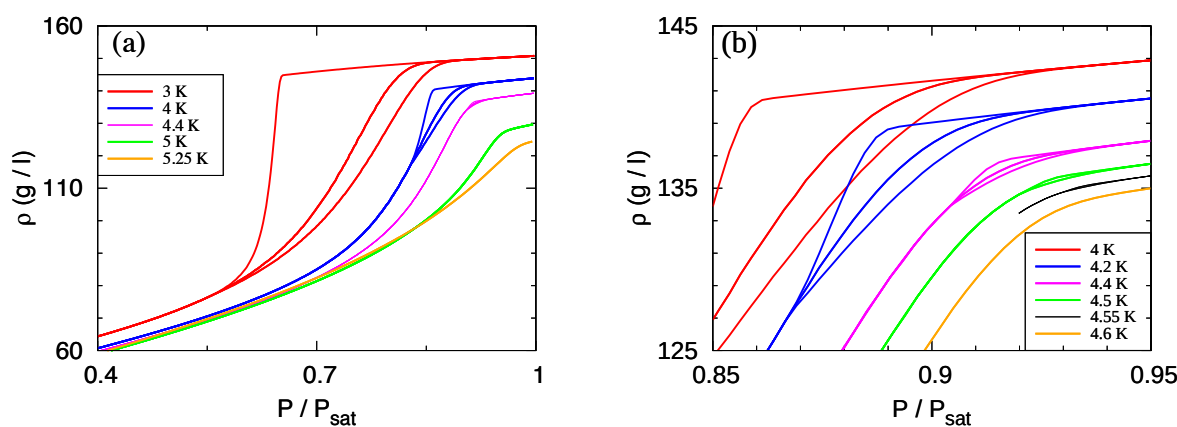


Figure 9: Predicted helium isotherms for a gaussian distribution of independent pores of average radius 4 nm and relative width 20%. For each temperature, the central branch corresponds to condensation (or evaporation) at equilibrium, and the external branches to cavitation or activated condensation, respectively. (a) Shape evolution from low temperatures to above T_c ; (b) Zoom around T_{ch} . For lisibility, the equilibrium branch is not shown at 4.5 K and above. The loop closes just above $T_{\text{ch}}=4.55$ K.

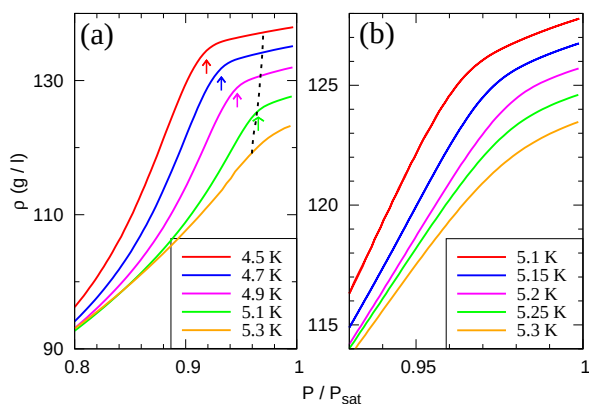


Figure 10: (a) Predicted helium isotherms above T_{ch} (same parameters as in Figure 9). Arrows indicate the equilibrium pressure for a pore diameter of 5.6 nm. The dashed line is the same as in Figure 8b, for a single pore of radius 4 nm; (b) Behaviour around T_c .

For independent pores distributed in diameter, we obtain the isotherm loop by averaging

the individual isotherms with a weight proportional to the pore radius distribution, $\Pi(R)$. For the sake of concreteness, we consider the same distribution as Saam and Cole, namely $\Pi(R) = \exp[-(R - R_{\text{av}})^2/(2\Delta^2)]$, with $R_{\text{av}}=4$ nm and $\Delta=0.2 R_{\text{av}}$. As seen in Figure 9a, this averaging suppresses the jumps of Figure 8b. For each temperature, Figure 9a shows three different branches. The central one corresponds to condensation (or evaporation) at equilibrium, while the right and left ones correspond to activated condensation or evaporation, respectively. Note that the activated evaporation (cavitation) branch can only be observed if the pores are closed by constrictions of small enough diameter and negligible volume. At some temperature T , all pores of radius smaller than $R_c(T)$ fill reversibly. The pressure at the lower closure point of the hysteresis loop is then $P_{\text{eq}}(R_c(T))$. As the temperature increases, more and more pores fill and empty reversibly, and the hysteresis loop decreases in size. On the scale of Figure 9b, it closes at $T_{\text{ch}} \approx 4.55$ K, about 10% below the bulk critical temperature. Note that T_{ch} is larger than 4 K, the critical temperature for $R_0 = R_{\text{av}}=4$ nm. This results from the diameter dependence of T_{ch} . Indeed, the observed critical temperature is controlled by the largest pores with a detectable contribution, in this case corresponding to a pore radius of 6 nm, about 2Δ larger than the average pore radius. Above the closure temperature, all pores fill or empty at equilibrium, and the isotherm shape is uniquely determined (Figure 10a). The density jump existing below T_c for a single pore is washed out by the averaging process. The kink-like feature visible above T_{ch} occurs at a lower pressure than the kink for a single pore of radius R_{av} . Its origin is indeed different, its position corresponding to the pressure above which most of the pores of the distribution are filled (compare to the arrows indicating the equilibrium pressure for a pore radius $R_0 + 2\Delta = 5.6$ nm). Finally, Figure 10b shows that the isotherm evolution around T_c presents no hint of the critical point at 5.195 K. Therefore, using porous systems with very well defined pores diameters is mandatory to detect a confinement induced shift of the critical point through isotherms measurements.

Below T_{ch} , the loop shape actually observed will depend on the nature of the conden-

sation and evaporation processes, which itself depends on the geometry. For perfect, non interconnected, cylindrical pores open at both ends, the condensation branch corresponds to activated condensation and the evaporation one to equilibrium. These two branches are roughly parallel, corresponding to the IUPAC H1 type.⁷⁷ This is the typical behaviour observed in silica MCM-41 templated pores.⁶⁰⁻⁶² In contrast, for a collection of ink bottles with cavities broadly distributed in diameter and narrow constrictions of negligible volume, condensation will take place at the equilibrium pressure of the cavities (the filled constrictions acting as germs), and evaporation at the cavitation pressure of the cavities (for temperatures large enough for the cavities to empty before the constrictions). In this case, the loop shape depends on temperature. At low temperature, it is close to triangular, corresponding to a H2 type in the IUPAC classification. While an H2 shape is often considered as a signature of coupling effects, our model implies that it can also be observed in the absence of such effects. It then results from the small dependence on radius of the cavitation pressure, as compared to that of the activated condensation pressure (Figure 5). This small dependence makes however the evaporation branch not strictly vertical. As the temperature increases, less pores contribute to the hysteretic loop, and the density difference between the filled and film states for these pores becomes smaller. This causes the evolution of the isotherm from triangular to rounded (Figure 9). Note that the evaporation branch becomes less steep when the hysteresis closure temperature is approached. This contradicts Morishige's assumption^{27,78} that a steepening of the evaporation isotherms with increasing temperature points to a cavitation scenario.

The temperature evolution from a triangular H2 to a rounded loop shown in Figure 9b is strikingly similar to the behaviour observed for helium confined in disordered Vycor glass, a material consisting of interconnected pores.²⁸ This is surprising as, in contrast to ordered networks of cavities such as SBA-16,¹⁸ there is no clear separation of scale between "cavities" and "constrictions" in Vycor. In that case, strong coupling effects are expected. Smaller sections of pores filled by thermal activation are indeed germs for neighbouring sections of

larger diameter, and may trigger the condensation of these larger sections at or close to their equilibrium pressure. Inversely, a wide section surrounded by narrower sections may empty above its cavitation pressure through a collective percolation mechanism starting from the surfaces,^{12–14,79} also giving rise to a H2-type loop. In order to quantify the effect of coupling, we studied a simplified model of Vycor using the present model for describing individual pores, and a lattice approach for describing the network effects. While this study, going beyond the scope of this paper, will be reported elsewhere, its results show that, in some temperature range below T_{ch} , the combination of thermal activation and coupling makes most pores to fill at their equilibrium pressure and empty at their cavitation pressure. This accounts for the close similarity of Figure 9 to the actual data of Ref. 28 around T_{ch} . In contrast, at lower temperatures, our study predicts that evaporation takes place by percolation, a mechanism previously demonstrated for hexane at room temperature in Ref. 80. This change in the evaporation regime is experimentally confirmed by light scattering measurements of the spatial correlations of the vapor distribution.²⁸ These measurements further reveal that the crossover between percolation at low temperature and cavitation closer to T_{ch} takes place over a finite temperature range. Such a smooth crossover would not be possible if the cavitation pressure would not depend on the pore diameter. In such a case, all pores would empty by cavitation if the cavitation pressure were larger than the percolation pressure, or by percolation in the opposite case. The experimental observation of a crossover thus supports our theoretical conclusion that confinement affects the cavitation pressure.

3.5 Extracting the pore diameter distribution from isotherms measurements

In this section, we evaluate how pore diameter distributions determined using the BJH method⁷ deviate from actual distributions, due to the effects of thermal activation and finite liquid compressibility. On this aim, starting from isotherms computed by our model including

these effects, we apply a BJH-like procedure to infer a distribution which we compare to the one injected into the model. As we will show, an analysis based on the assumption of condensation at equilibrium yields a reasonable estimate of the pores diameter distribution in a wide temperature range around T_{ch} .

We first discuss the case of an assembly of independent, open at both ends, perfectly cylindrical pores. Ideally, this situation is expected to apply to porous alumina membranes. Previous experimental results on such systems with large pores (several tens of nanometers in diameter) have been analyzed assuming that condensation (resp. evaporation) takes place at the spinodal (resp. equilibrium) pressure, and using the Kelvin equation for a cylindrical or hemispherical meniscus (depending on which branch of the isotherm is considered) to convert pressure into pore radius.^{57,64} This is justified as long as the pore radius is much larger than R^* . In the opposite situation (alumina membranes with smaller pores, MCM-41's), the film should be taken into account. This has two consequences. First, the mass increment between two neighboring pressures along the isotherm does not only come from those pores which fill or empty in this pressure interval, but also from the pressure dependence of the film thickness. The standard BJH method consists in analyzing the isotherm backwards, starting from the largest pores, and recursively determining the pore radius distribution, assuming that, within the film, the fluid density has its bulk value, and estimating the pressure dependence of the film thickness from experimental data on flat surfaces. Second, the finite film thickness modifies the relationship between pressure and radius, as compared to the Kelvin equation. The BJH approach assumes that, on the evaporation branch, the equilibrium pressure for a pore of radius R_0 is given by the Kelvin equation for a pore of radius $R_0 - t$, where $t(P)$ is the pressure dependent film thickness. As shown by Derjaguin,⁴ this is not exact, and a better approximation for a van der Waals interaction is $R_0 - 3/2t$. Alternatively, a generalization of the BJH method consists in applying a similar recursive scheme, now using Saam and Cole's results to convert pressures into radii and to evaluate the pressure dependence of the film thickness. Such an approach however neglects the influence

of thermal activation.

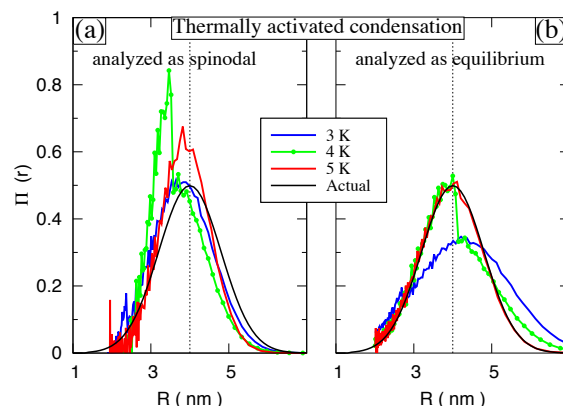


Figure 11: Evaluation of a BJH-like inversion scheme based on Saam and Cole's predictions. Pore diameter distributions $\Pi(R)$ deduced from the condensation branch of isotherms computed at 3, 4, and 5 K using the thermally activated model, $\alpha=1200 \text{ K}/\text{\AA}^3$ and a gaussian pore radius distribution centered on 4 nm (black line) ;(a): when assuming condensation at the spinodal; (b) when assuming condensation at equilibrium.

In order to evaluate the impact of this approximation, we applied the above scheme to condensation isotherms computed taking into account the effect of thermal activation (for the same gaussian pore radius distribution as above). Figure 11 shows the deduced pore diameter distribution for different temperatures, assuming respectively that the condensation takes place at the spinodal or at the equilibrium pressure, and neglecting the effect of the finite compressibility. At 3 K, the distribution recovered under the first assumption is nearly correct. This results from the fact that thermal activation is weak enough for condensation to take place near the spinodal pressure for nearly all pores. At 5 K, above T_{ch} , the second assumption, as expected, yields the correct distribution. At 4 K, none of the assumptions is exact, although the equilibrium hypothesis gives better results than the spinodal one. For this temperature, the pressure for activated condensation indeed lies halfway between the equilibrium and the spinodal pressure for the larger pores and is equal to the equilibrium pressure for the smaller (see Figure 6). We can draw two conclusions from this analysis: (i) applying the BJH approach to the condensation branch using the hypothesis of spinodal condensation is only correct at temperatures much smaller than T_{ch} ; (ii) in contrast, not too

far below the hysteresis closure temperature, and above, using the equilibrium pressure for the BJH analysis of the condensation branch yields a reasonable approximation of the pore diameter distribution (see Supporting Information for details).

Let us now consider the case of complex porous materials with coupled pores. As discussed in the introduction, this situation can be dealt by combining the present analytical model for individual pores with a numerical lattice approach taking into account the coupling between neighboring pores. Using this approach, we find that most of the pores fill at (or close to) their equilibrium pressure. In this case, an analysis of the condensation isotherms based on the hypothesis that pores fill independently at equilibrium recovers the correct pore distribution. As a consequence, for such coupled systems, we expect that a BJH-like analysis based on the single pore condensation isotherms computed within our model can be applied in a wide temperature range around T_{ch} .

For the specific case of temperatures close to T_c , one might be concerned by the influence of the large liquid compressibility. Indeed, in this situation, filled pores contribute to the mass increment between two neighboring pressures along the isotherm. We show in the Supporting Information how an iteration of the BJH procedure can be used to take this change into account. Notwithstanding, distributions computed ignoring this effect reasonably agree with the injected distribution showing that, in most cases, it can safely be ignored.

A more serious problem is the sensitivity of the pore diameter determination to the choice of the van der Waals interaction strength, which is not a priori known. To demonstrate this sensitivity, we analyze in Figure 12 isotherms computed with a given interaction strength using the modified BJH scheme with three different values of the interaction. The original distribution is recovered when using the correct interaction strength, thus checking the efficiency of the inversion method. As expected, underestimating (rep. overestimating) the attraction strength leads to a distribution shifted toward lower (resp. larger) R values. For the larger temperature (5 K), the distribution peaks close to 3 and 5 nm, the radii R such that the on-axis potential α/R^3 is a constant. This shows that the interaction effect domi-

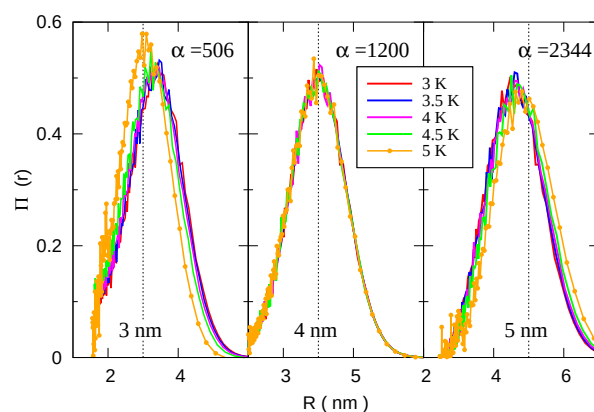


Figure 12: Sensitivity of the BJH-like inversion scheme to the choice of the van der Waals interaction α . Pore radius distribution $\Pi(R)$ deduced from the equilibrium branch of isotherms computed at 3, 3.5, 4, 4.5 and 5 K for $\alpha=1200$ K/Å³ and a gaussian pore radius distribution centered on 4 nm. The three panels correspond to applying the inversion scheme using three different values for α .

notes the capillary effect at this temperature. At lower temperatures, the distributions are less shifted, because the increased capillary effects reduce the relative influence of interaction. A correct estimate of the van der Waals parameter, hence of the average pore radius, can be found by requiring that the inferred distribution does not depend on temperature. However, this procedure requires to analyze isotherms over an extended temperature range (60%-96% T_c in the present case). The correct pore distribution, nor the average pore radius, cannot be deduced from isotherms measurements restricted to high or low temperatures only. Referring to Figure 2, selecting the proper value of α requires to cover a temperature range large enough to reveal the curvature of R_K/R_0 versus R^*/R_0 in a log-log plot. Otherwise, a change in the interaction strength can always be compensated for by a (temperature independent) change of R_0 . The distributions for different temperatures then coincide, whatever the choice of α . If this is the case, it is necessary to estimate α by a different method. As shown in the Supporting Information, comparing the experimental pressure dependence of the lower point of the closure loop to the behaviour predicted by our model provides such an estimate. We estimate that this procedure should allow to determine the average pore radius within 20%.

4 Conclusions

In this paper, we have used a semi-macroscopic approach to determine the effect of thermal activation on the condensation and evaporation of a fluid confined in a perfect cylindrical nanopore. If the fluid compressibility is neglected, the energy barrier, derived assuming a van der Waals interaction between the fluid and the substrate, can be expressed in terms of two adimensional parameters, measuring respectively the attraction strength and the undersaturation. This approximation holds up to temperatures very close to the critical point. For a given fluid, this allows to compute the temperature dependence of the critical pressures for activated condensation or evaporation (cavitation), respectively. These pressures merge with the equilibrium pressure previously obtained by Saam and Cole above a critical temperature T_{ch} smaller than the bulk critical temperature T_c . Our model thus confirms and quantifies the idea first put forward by Machin³² that thermal activation alone can account for the closure of the hysteresis loop below T_c , without any modification of the fluid surface tension by the confinement. It also predicts that decreasing the pore diameter, *i.e.* increasing the confinement, decreases both the cavitation pressure and T_{ch} . We next computed the isotherm shape for assemblies of cylindrical pores with distributed diameters. When these pores are both interconnected and distributed in diameter, evaporation can proceed by cavitation. In this case, the diameter dependence of the cavitation pressure and of T_{ch} causes an evolution from a H2 behaviour at low temperature to a H1-like behaviour below the closure temperature T_{ch} . We also evidenced the sensitivity of the isotherm shape to the liquid compressibility, due to the effect of the van der Waals attraction on the local liquid density.

Our physical approach reproduces a number of results previously obtained using sophisticated microscopic numerical methods, such as the dependence of the energy barrier on undersaturation,³⁷ the existence of a critical temperature for hysteresis depending on the pore diameter,¹⁹ or the evolution of the loop shape with increasing temperature in a numerical model of Vycor.¹⁶ This gives us confidence that our *generic* semi-macroscopic model

properly describes condensation and evaporation in not too small nanopores. As such, it can be used to design optimal ink-bottles to observe and study cavitation in a confined geometry, or to analyze the diameters of constrictions in ordered materials with cagelike mesopores. It can also be used as a building block in a multiscale approach, allowing to study disordered porous materials, such as 1D chains, as a modelisation of corrugated pores, or 3D networks, as a modelisation of Vycor and other porous glasses. Using our predictions for the diameter dependence of the evaporation and condensation pressures, one can determine whether, depending on the network geometry or the temperature, the pores of such systems mostly fill at equilibrium. If this condition is fulfilled, the radius dependence of the equilibrium pressure determined by Saam and Cole is a good basis for determining the pore diameter distribution, provided that the van der Waals interaction is known from isotherms measurements performed over a wide enough temperature range. In summary, our work should provide an efficient tool to analyze pore diameter distributions of porous materials.

Acknowledgement

This paper is dedicated to the memory of our late colleague L. Puech, with whom these studies were started. The authors thank J. Beamish, V. Doebele, P. Spathis, and J. C. Anglès d'Auriac for their useful suggestions and/or critical reading. They also thank one of the referees for having pointed to them Refs. 58 and 48. They acknowledge ANR for initial funding (ANR-06-BLAN-0098).

Supporting Information Available

The free of charge Supporting Information contains a pdf file where we derive in detail the effect of thermal activation for condensation and evaporation in a single pore. We also justify why the liquid compressibility can be neglected when calculating the characteristic pressures for condensation of evaporation. Several points of our paper are also developed,

in particular the influence of several factors on the determination of the pore distribution through the BJH-like method. Finally, we detail the physical data used in our paper.

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equation. Here, we use a parabolic extrapolation. See Supporting Information for details.

- (75) The supercritical isotherm is found by averaging the density $\rho(P(r))$ where the local pressure $P(r)$ is given by $\mu(P(r)) + U(r) = \mu_g(P_g^\infty)$, and μ is obtained by integrating the bulk $P - \rho$ behaviour given by Ref. 73.
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Graphical TOC Entry

